



Project Report

Computer Networks(CSC-502)

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Semester:

CS-5th (Regular)

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Design and Implementation of a Secure Multi-Floor Network for VIC Modern Hotel

Abstract

This project focuses on the design and implementation of a secure, scalable, and efficient computer network for a fictional multi-floor organization named **VIC Modern Hotel**. The network integrates multiple networking concepts including VLAN segmentation, inter-VLAN routing, dynamic IP addressing using DHCP, OSPF routing protocol, wireless networking, SSH remote access, and switch port security. The aim of this project is to simulate a real-world enterprise hotel network that supports departmental isolation, secure communication, wireless access, and centralized management across multiple floors.

1. Introduction

Modern hotels require reliable and secure networks to support daily operations, guest services, and internal management. Departments such as reception, finance, IT, and administration must communicate efficiently while maintaining logical separation and security.

This project demonstrates the application of enterprise-level networking technologies in a hotel environment. The VIC Modern Hotel network is designed to support multiple floors, wired and wireless devices, printers, and secure remote access while ensuring network efficiency and reliability.

2. Project Objectives

The primary objectives of this project are:

- To design a multi-floor enterprise network for a hotel environment
- To implement VLAN-based departmental segmentation
- To enable inter-VLAN communication using router subinterfaces
- To configure OSPF for dynamic routing
- To provide dynamic IP addressing using DHCP
- To integrate secure wireless networking
- To implement network security using SSH and port security
- To test and verify complete network functionality

3. Network Overview

The VIC Modern Hotel consists of three floors, each containing multiple departments. Each floor is equipped with one router and one switch. Routers are interconnected using serial DCE links to enable communication between floors. All departments are logically separated using VLANs.

3.1 Floor Structure

- **First Floor:** Reception, Store, Logistics
- **Second Floor:** Finance, HR, Sales & Marketing
- **Third Floor:** IT, Admin

Each department contains PCs and printers, while wireless access is provided for mobile devices.

4. Network Design Architecture

The network follows a hierarchical and modular design consisting of three routers (F1, F2, and F3), one switch per floor, serial DCE connections between routers, and access points for wireless connectivity. This design improves scalability, manageability, and fault tolerance.

5. VLAN and IP Addressing Scheme

Each department is assigned a unique VLAN ID and subnet to reduce broadcast traffic and enhance security.

Floor	Department	VLAN ID	IP Network
1	Reception	80	192.168.8.0 /24
1	Store	70	192.168.7.0 /24
1	Logistics	60	192.168.6.0 /24
2	Finance	50	192.168.5.0 /24
2	HR	30	192.168.3.0 /24
2	Sales & Marketing	40	192.168.4.0 /24
3	IT	10	192.168.1.0 /24
3	Admin	20	192.168.2.0 /24

6. Inter-VLAN Routing

Inter-VLAN routing is implemented using the Router-on-a-Stick approach. Router subinterfaces are configured with IEEE 802.1Q encapsulation, and each subinterface acts as the default gateway for its respective VLAN, allowing controlled communication between departments.

7. Router Interconnection and IP Addressing

Routers are interconnected using serial links with /30 subnets to efficiently utilize IP addresses.

Router Link	Subnet	IP Assignment
F1 – F2	10.10.10.4 /30	F1: 10.10.10.5, F2: 10.10.10.6
F1 – F3	10.10.10.8 /30	F1: 10.10.10.9, F3: 10.10.10.10
F2 – F3	10.10.10.0 /30	F2: 10.10.10.1, F3: 10.10.10.2

Clock rates are configured on all DCE interfaces.

8. OSPF Routing Configuration

OSPF is configured on all routers under Area 0. This allows dynamic exchange of routing information and ensures fast and reliable communication between all VLANs and floors.

9. DHCP Configuration

Routers are configured as DHCP servers with separate pools for each VLAN. Each pool specifies the network address, default gateway, and DNS server, enabling automatic IP assignment for all devices.

10. Wireless Network Implementation

Wireless access points are deployed on each floor with unique SSIDs and password protection. Laptops, tablets, and smartphones connect successfully and receive IP addresses via DHCP.

11. Security Implementation

11.1 SSH Configuration

SSH is configured on all routers using hostname, domain name, user authentication, and RSA key generation. VTY lines are restricted to SSH-only access.

11.2 Port Security

Port security is applied to the IT department switch port using sticky MAC addressing, allowing only one authorized device. Violation mode is set to shutdown to block unauthorized access.

12. Testing and Verification

Successful testing includes DHCP IP assignment, inter-VLAN ping tests, OSPF route verification, SSH remote login testing, wireless connectivity, and port security enforcement.

13. Conclusion

This project successfully implements a secure and scalable enterprise network for VIC Modern Hotel. It demonstrates real-world networking concepts including VLANs, routing, DHCP, wireless networking, and security. The project serves as an excellent practical model for enterprise Cisco networking.

14. Future Enhancements

Future improvements may include firewall implementation, ACLs, centralized wireless management, SNMP monitoring, and redundant link configurations.